Yoonjae Baek

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Education

M.S. **Pohang University of Science and Technology (POSTECH)**, Artificial Intelligence — Pohang, South Korea
- Graduate School of Artificial Intelligence, Computer Vision Lab — Feb 2025 – present
- Advisor: Prof. Suha Kwak

B.S. **Sungkyunkwan University (SKKU)**, Computer Education — Seoul, South Korea
- Advisor: Prof. Sungeun Hong — Mar 2018 – Feb 2025

Publications

Taming VLAs under Robot Execution Errors: Self-Compensation and Stress Testing — *Under Review* 2026
Sohyun Lee*, **Yoonjae Baek***, Jaesang Won, Jinnyeong Kim, Hyunwoo Kang, Seung-Hwan Baek, Ivan Laptev, Suha Kwak
*equal contribution

When to Switch: Reliable Action-Chunk Extension for Vision-Language-Action Models — *Under Review* 2026
Seonghoon Yu, Dongwon Kim, HyungRok Jung, **Yoonjae Baek**, Byung-kwan Lee, Suha Kwak, Jeany Son

EventCoT: Event-centric Video Chain-of-thought for Reasoning Temporal Localization — *Under Review* 2026
Youngkil Song, **Yoonjae Baek**, Dongwon Kim, Inho Kim, Dongkeun Kim, Suha Kwak

Dual Path Semantic Segmentation with Locality Guidance Tokens — *Neurocomputing* 2026
Jaeik Kim, **Yoonjae Baek**, Aecheon Jung, Hongyeob Kim, Soyun Choi, Youngeun Kim, Jaeyoung Do, Sungeun Hong

Experience

Computer Vision Lab, POSTECH, Research and Teaching Assistant — Pohang, South Korea, Feb 2025 – present

AI & Media (AIM) Lab, Sungkyunkwan University, Undergraduate Research Intern — Seoul, South Korea
- Advisor: Prof. Sungeun Hong — Jan 2024 – June 2024

Dept. of Computer Education Student Council, SKKU, President — Seoul, South Korea, Nov 2019 – Nov 2020

Honors and Awards

Gold Prize, 38th Workshop of Image Processing and Image Understanding (IPIU) — Feb 2026
Jeju, South Korea

Services

Student Volunteer, Conference on Robot Learning (CoRL) — 2025

Interests

Research Interests: Embodied AI, Vision-Language-Action Models, Real-World Deployment, Video Understanding